130

# Bladder and Female Urethral Diverticula

*Lindsey Cox, and Eric S. Rovner*

## Keywords

bladder diverticulum; diverticulectomy; dysuria; neurogenic bladder; pelvic pain; urethral diverticulum; urinary incontinence; urinary retention; urinary tract infection; voiding dysfunction

## Questions

1. Congenital bladder diverticula are:
    a. usually multiple.
    b. strongly associated with bladder outlet obstruction.
    c. often found in smooth-walled bladders.
    d. located at the dome.
    e. usually less than 1 cm in size.
2. Acquired bladder diverticula are most commonly located:
    a. near the urethrovesical junction.
    b. adjacent to the ureter.
    c. at the dome.
    d. at the 10 and 2 o'clock positions.
    e. posteriorly.
3. Videourodynamic evaluation in an adult female with a bladder diverticulum will likely reveal:
    a. impaired compliance.

b. bladder outlet obstruction.
c. intrinsic sphincter deficiency (ISD).
d. low-pressure, low-flow voiding.
e. no abnormality.

4. Pathologic examination of a surgical bladder diverticulectomy specimen will likely reveal:
   a. absence of epithelium.
   b. premalignant or malignant changes.
   c. nephrogenic metaplasia.
   d. a poorly developed muscularis propria layer.
   e. trabeculation of the smooth muscle layer.
5. The most common malignant tumor associated with bladder diverticula is:
   a. urothelial.
   b. squamous cell.
   c. adenocarcinoma.
   d. sarcomatous.
   e. undifferentiated.
6. A 68-year-old man presents with hematuria. Cystoscopy reveals a 15-cm bladder diverticulum with a 3-mm papillary lesion at the base of the diverticulum. The next step is:
   a. biopsy of the papillary lesion.
   b. transurethral resection of the papillary lesion with deep muscle resection.
   c. urodynamics and transurethral prostatectomy (TURP) if bladder outlet obstruction is noted.
   d. bladder diverticulectomy.
   e. radical cystectomy and urinary diversion.
7. Acquired bladder diverticula are commonly found in association with:
   a. prostatic obstruction.
   b. calyceal diverticula.
   c. nephrogenic adenoma.
   d. infection of perivesical glands.
   e. erectile dysfunction (ED).

8. A 65-year-old man with bladder outlet obstruction and a 5-cm bladder diverticulum undergoes uneventful TURP. Postoperatively, the patient's symptoms are resolved, and a voiding cystourethrogram (VCUG) demonstrates satisfactory emptying of the bladder and the bladder diverticulum. The next step is:
    a. annual surveillance with cystoscopy and urine cytology.
    b. discharge from urologic care.
    c. transvesical bladder diverticulectomy.
    d. repeat urodynamics.
    e. computed tomographic (CT) cystogram.
9. Ten years following TURP, a 71-year-old man with congestive heart failure (CHF) and atrial fibrillation who is on warfarin sodium (Coumadin) has recurrent urinary tract infections (UTIs) and an American Urological Association (AUA) symptom score of 25. A videourodynamic study shows a 14-cm poorly emptying bladder diverticulum. The peak subtracted detrusor pressure (Pdet) during micturition is 15 cm $H_2O$ with a Qmax of 3 mL/s. Renal ultrasonography is normal. The next step is:
    a. repeat TURP.
    b. clean intermittent self-catheterization (CIC).
    c. observation.
    d. bethanechol.
    e. CT urography.
10. Endoscopic examination of the lower urinary tract in the setting of bladder diverticula:
    a. is best performed with a rigid cystoscope.
    b. is associated with a high risk of perforation.
    c. should include examination of the entire interior of the diverticulum.
    d. is not indicated if an elective submucosal bladder diverticulectomy is planned.
    e. should always be performed with concomitant bilateral retrograde pyelography (RPG).
11. Bladder diverticula:
    a. often do not produce specific symptoms.

b. can be associated with UTIs.
c. are commonly diagnosed incidentally during the evaluation of other symptoms or conditions.
d. may be associated with persistent pyuria.
e. all of the above.

12. Bladder diverticula associated with bladder outlet obstruction:
    a. are usually found in the absence of cellules and saccules.
    b. are associated with the universal finding of ipsilateral vesicoureteral reflux.
    c. cannot be imaged by CT.
    d. are associated with medial deviation of the pelvic ureter.
    e. are less likely to be associated with malignancy compared with congenital bladder diverticula.
13. Increased size of urethral diverticula (UDs) at presentation correlates with:
    a. increased symptoms.
    b. risk of UTI.
    c. risk of recurrence postoperatively.
    d. risk of malignancy.
    e. risk of incontinence.
14. Common symptoms associated with UDs include all of the following EXCEPT:
    a. vaginal pruritus.
    b. dysuria.
    c. dyspareunia.
    d. postvoid dribbling.
    e. urinary urgency and frequency.
15. In a 35-year-old woman a 1.5-cm firm anterior vaginal wall mass is noted approximately 2 cm proximal to the urethral meatus at the level of the mid-urethra without distorting the urethral meatus. It is nontender. Urine analysis is unremarkable. This mass may represent any of the following EXCEPT:
    a. vaginal wall cyst.
    b. Skene gland abscess.
    c. UD.
    d. vaginal leiomyoma.

e. Gartner duct cyst.

16. The ostium of a UD is:
    a. most commonly found in the proximal third of the urethra.
    b. most commonly found at the 10 and 2 o'clock position in the urethral lumen.
    c. usually seen on transvaginal ultrasound imaging.
    d. unable to be visualized with rigid cystoscopy.
    e. usually located in the ventrolateral urethra.
17. Two weeks after removal of a 5-cm proximal UD extending beneath the trigone of the bladder, a 48-year-old woman returns to the office with complaints of urine staining her undergarments. Possible etiologies include:
    a. urethrovaginal fistula.
    b. ureterovaginal fistula.
    c. vesicovaginal fistula.
    d. stress urinary incontinence (SUI).
    e. all of the above.
18. During excision of the epithelial lining (sac) of a UD, a portion of the indwelling urethral catheter is seen at the base of the dissection. The next step is to:
    a. close the urethra and abort the procedure.
    b. perform buccal mucosal urethroplasty and abort the procedure.
    c. complete the urethral diverticulectomy.
    d. construct a vaginal inversion flap and close the urethra.
    e. close the urethra primarily, place a suprapubic tube, and harvest a Martius flap.
19. A VCUG is performed for evaluation of a possible UD. The filling images are nondiagnostic. The radiologist calls you because the patient is unable to void under fluoroscopy in the radiology suite. The patient is taken off the imaging table and is able to void in the adjacent bathroom. The next step is to:
    a. obtain a CT cystogram.
    b. do a transvaginal ultrasound.
    c. obtain postvoid images.
    d. implement endoluminal magnetic resonance imaging (MRI).

e. positive pressure urethrography (PPU).

20. The most common malignancy found in UDs is:
    a. squamous.
    b. urothelial.
    c. adenocarcinoma.
    d. undifferentiated.
    e. sarcomatous.
21. Principles of surgical urethral diverticulectomy include all of the following EXCEPT:
    a. preservation or creation of urinary continence.
    b. excision of all identifiable periurethral fascia.
    c. identification of the ostium of the UD.
    d. closure of periurethral fascia following removal of the UD.
    e. watertight closure of the urethra.
22. The initial event implicated in the formation of most UDs is:
    a. congenital lack of fusion of the urethral crest.
    b. infection of vaginal cysts.
    c. a traumatic vaginal delivery.
    d. infection of the periurethral glands.
    e. dysfunctional voiding.
23. In a patient with bothersome SUI and UD, anti-incontinence surgery is being considered. Of the choices listed below, the best concomitant surgical procedure to treat the SUI is to provide a:
    a. transobturator midurethral sling.
    b. retropubic midurethral sling.
    c. single-incision synthetic sling.
    d. autologous pubovaginal fascial sling.
    e. polypropylene bladder neck sling.
24. The ostia of UD are most commonly found at:
    a. 12 o’clock.
    b. 6 o’clock.
    c. 4 and 8 o’clock.
    d. 10 and 12 o’clock.
    e. none of the above.

# Imaging

1. A 47-year-old woman presents with dribbling and recurrent UTIs; see Fig. 130.1.

The most likely diagnosis on this axial and coronal T2-weighted MRI done with an endovaginal coil is:

a. bladder prolapse.
b. urethral diverticulum.
c. bladder diverticulum.
d. ureteral duplication.
e. ectopic ureter.

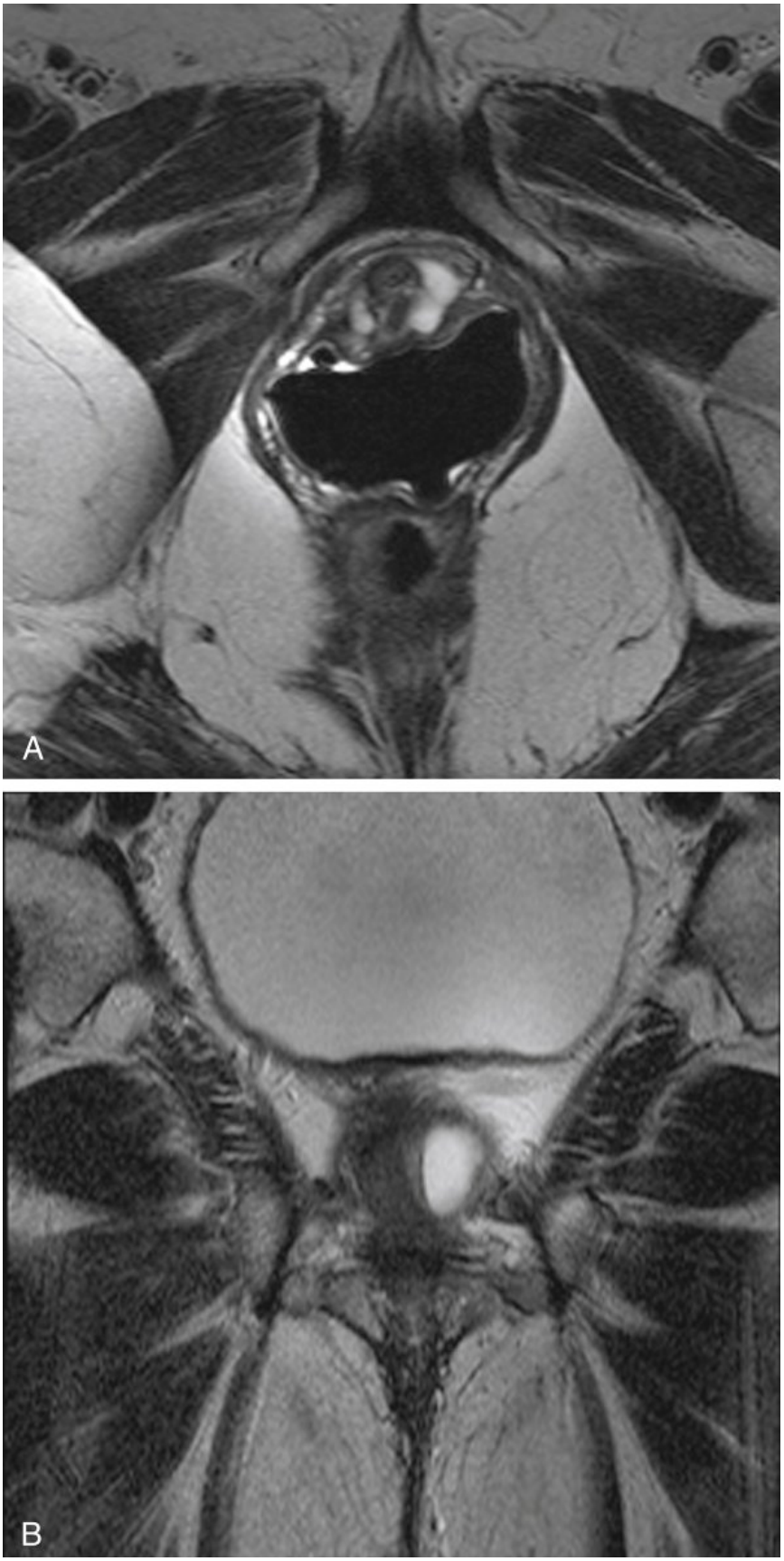

A
B

**FIG. 130.1** Endovaginal T2 MRI images. (A) Axial. (B) Coronal.

## Answers

1. **c. Often found in smooth-walled bladders.** Unlike acquired bladder diverticula, congenital lesions are found in smooth-walled bladders, often adjacent to the ureteric orifice. **Congenital bladder diverticula are usually solitary, often large, and not associated with bladder outlet obstruction.** Congenital diverticula at the dome may be related to prune-belly syndrome, posterior urethral valves, or urachal anomalies.
2. **b. Adjacent to the ureter.** Similar to congenital bladder diverticula, acquired bladder diverticula are often located near the ureteric orifice. This area of the bladder is thought to be a location of relative anatomic weakness of the bladder wall, predisposing to the formation of bladder diverticula.
3. **b. Bladder outlet obstruction. Bladder diverticula in adults are most commonly associated with some type of bladder outlet obstruction.** In an adult female this may be due to prior anti-incontinence surgery (iatrogenic), dysfunctional voiding, neurogenic causes (e.g., detrusor external sphincter dyssynergia), or a variety of other conditions such as obstructing vaginal or urethral masses (e.g., malignancy).
4. **d. A poorly developed muscularis propria layer.** Although varying amounts of detrusor muscle fibers may be found on pathologic examination of surgical excised bladder diverticula, the muscularis propria layer is usually incomplete and the fibers are disorganized. Such a lack of a well-developed muscularis propria layer is a hallmark of a bladder diverticulum. Absence of an epithelial layer, nephrogenic metaplasia, and malignant changes are uncommonly reported in the histologic evaluation of bladder diverticula. Typically there is no trabeculation within the wall of a bladder diverticulum.

5. **a. Urothelial.** The most common malignant tumors seen within bladder diverticula are urothelial. The other types of tumors can be seen but are much less common.
6. **a. Biopsy of the papillary lesion.** A papillary lesion in a bladder diverticulum, similar to the rest of the urinary tract, may indicate a malignancy, but it may also represent a benign lesion. Biopsy is warranted. Care should be taken to avoid perforation of the diverticular wall during the biopsy as the wall lacks a muscularis propria layer. A 3-mm lesion can easily be biopsied without the use of a resectoscope. In the event that the biopsy demonstrates malignancy, perforation of the bladder wall risks malignant dissemination.
7. **a. Prostatic obstruction.** Acquired bladder diverticula are most commonly associated with bladder outlet obstruction (approximately 70%). They are also more common in males than in females. The most common cause of bladder outlet obstruction in males is prostatic obstruction. There is no known association of bladder diverticula with calyceal diverticula, ED, nephrogenic adenoma, or perivesical gland infection.
8. **a. Annual surveillance with cystoscopy and urine cytology.** Although the bladder diverticulum drains well and the patient is asymptomatic, long-term follow-up of bladder diverticula is warranted. **The natural history of such lesions is unknown, and although the etiology of malignant transformation is thought to involve urinary stasis, this is unproved. Furthermore, it is possible that premalignant changes may have occurred during the time that the diverticulum was not draining well.**
9. **b. Clean intermittent self-catheterization (CIC).** This patient is at high risk for major abdominal surgery. Removal of his diverticulum is likely to be a large surgical undertaking with substantial risk and would be ill advised unless other conservative measures failed. Endoscopic treatment with TURP is also a risk because of anticoagulation. His detrusor pressure is low, suggesting that even an adequate outlet procedure (in this patient without definite evidence of obstruction) will not provide satisfactory emptying of the diverticulum. He has normal upper tracts but recurrent UTIs,

thus observation would not be optimal. It is likely that his UTIs are due to urinary stasis in the diverticulum. CIC should adequately empty the diverticulum and reduce the risk of ongoing UTIs.

10. **c. Should include examination of the entire interior of the diverticulum.** Periodic endoscopic examination of bladder diverticula is warranted because of the risk of malignant transformation and early transmural involvement. The entire lining of the diverticulum should be examined. Rigid and flexible cystoscopes can be used. The upper urinary tract should be imaged. However, RPGs are not necessary unless otherwise indicated for another reason (e.g., a filling defect on urography).
11. **e. All of the above.** Bladder diverticula are most commonly recognized incidentally on evaluation for other signs and symptoms. There are no symptoms that are specific to bladder diverticula. Bladder diverticula are associated with UTIs and pyuria in some individuals.
12. **d. Are associated with medial deviation of the pelvic ureter.** Bladder diverticula found in association with lower urinary tract obstruction are commonly seen with saccules and cellules. **Reflux is not a common finding with acquired bladder diverticula but may be present in some individuals, especially those with "Hutch" diverticula.** Congenital bladder diverticula are not associated with malignancy. Medial deviation of the ureter can be seen on intravenous urography and CT, with some large bladder diverticula due to the location of the diverticula relative to the ureter.
13. **c. Risk of recurrence postoperatively.** Although some UDs can be quite large, symptoms, malignancy risk, incontinence, and risk of UTI are not known to correlate with size. Large diverticula, including those extending in a "saddle bag" configuration, are more likely to recur.
14. **a. Vaginal pruritus.** Vaginal pruritus is not a symptom of UDs. Inflammatory and infectious conditions as well as lichen sclerosis (vulvar dystrophy) may cause vaginal pruritus. The other symptoms listed are often, although not invariably, individually or collectively associated with UDs. The classic triad of symptoms of

UDs includes the "3 D's"—dysuria, postvoid dribbling, and dyspareunia—although these are seen only rarely in the same patient.

15. **b. Skene gland abscess. Skene gland abscess is usually associated with distortion of the urethral meatus due to its distal location.** These lesions are also often symptomatic, with associated dyspareunia and tenderness on physical examination. **Vaginal wall cysts, vaginal leiomyomas, and Gartner duct cysts are firm anterior vaginal wall masses and are often nontender.** UDs in some cases may be asymptomatic and found incidentally in the evaluation of other conditions, but they do not result in distortion of the urethral meatus.
16. **e. Usually located in the ventrolateral urethra. The ostium (opening) of a UD is most commonly found in the middle or distal third of the urethra, at the 4 and 8 o'clock positions.** They are usually too small to be visualized by ultrasound but can often be seen with both rigid and flexible cystoscopy as small openings in the urethral lumen ventrolaterally.
17. **e. All of the above.** Complications of urethral diverticulectomy include urinary fistulas as well as stress incontinence. Large diverticula may extend beneath the trigone of the bladder, and excision of such lesions risks injury to the urethra, bladder, and ureters. De novo SUI may occur following urethral diverticulectomy, which may be due to distortion or injury to the sphincter mechanism.
18. **c. Complete the urethral diverticulectomy.** Successful excision of a UD involves removal of the ostium that connects with the urethral lumen. This often results in direct visualization of the urethral catheter within the urethral lumen during surgery. The urethral defect is closed primarily with absorbable suture in a watertight fashion following completion of the removal of the sac. Additional procedures such as buccal mucosal urethroplasty, Martius flap, or vaginal flaps are not necessary to close the urethra.
19. **c. Obtain postvoid images.** In the absence of voiding images, many UDs will not be visualized during fluoroscopy as there is no contrast in the urethra. This is a nondiagnostic but incomplete

study. Postvoid images will often reveal retained contrast in the diverticulum. CT, ultrasound, MRI, and PPU are all potentially useful studies in the evaluation of diverticula, but every effort should be made to maximize the diagnostic potential of each radiographic imaging technique, and a postvoid image in this clinical scenario may provide useful diagnostic information.

20. **c. Adenocarcinoma.** The most common malignant tumor type found in UDs is adenocarcinoma. Although the other malignant tumors listed may be found, adenocarcinoma is the most common.
21. **b. Excision of all identifiable periurethral fascia. Preservation of periurethral fascia is an important step in urethral diverticulectomy. This tissue is very important in reconstruction of the urethra to prevent fistula formation and close dead space.** This tissue should not be excised.
22. **d. Infection of the periurethral glands.** Infection of the periurethral glands is felt to be the initial step in the formation of UDs. Such infection may lead to periurethral abscess formation and development of a cavity or space within the periurethral fascia that then becomes the anatomic location of a UD.
23. **d. Autologous pubovaginal fascial sling.** SUI may accompany UDs. In symptomatic patients, concomitant repair can be considered. UDs connect to the urethral lumen, and therefore the surgical excision of these lesions requires suture repair and closure of the ostium of the UD where it connects to the urethra. **In such settings, the use of synthetic material for a sling is not advisable because of the risk of subsequent erosion of the synthetic material into the urethra.** Autologous fascia is an excellent choice for a concomitant sling at the time of urethral diverticulectomy.
24. **c. 4 and 8 o'clock. UDs are thought to originate from infection of the periurethral glands. Such glands are located along the urethra and arborize laterally. They generally drain medially into the urethral lumen ventrolaterally at approximately the 4 and 8 o'clock positions of the middle and distal third of the urethra.**

## Imaging

1. **b. Urethral diverticulum.** These images demonstrate a fluid collection surrounding the urethra, compatible with a saddlebag UD. The coronal image clearly shows that the collection is separate from the bladder. (Options a and c are incorrect. An ectopic ureter would not have a saddlebag configuration.)

## Chapter Review

1. When the diverticulum encompasses the ureteral orifice in the setting of neurogenic bladder and vesicoureteral reflux, it is termed a *Hutch diverticulum.*
2. Congenital diverticula generally occur lateral and posterior to the ureteral orifice and are often associated with vesicoureteral reflux.
3. Acquired bladder diverticula usually occur in the setting of obstruction or neurogenic vesicular dysfunction.
4. The major complications of diverticula include recurrent UTIs, stones, carcinoma or premalignant change in the diverticulum, and upper tract deterioration as a consequence of obstruction or reflux.
5. Many diverticula are located adjacent to the ureter and may be very adherent to it. This has implications in surgical resection.
6. The urethropelvic ligament in the female is composed of two parts: (1) endopelvic fascia and (2) periurethral fascia. Within these two leaves of fascia lie the urethra, and this is the location of most UDs in women.
7. The etiology of UDs in women has been attributed to recurrent UTI of periurethral glands with obstruction, suburethral abscess formation, and subsequent rupture of the infected gland into the urethra.
8. Skene glands do not communicate with the urethra.
9. Gartner duct cysts are located on the anterolateral vaginal wall from cervix to introitus.
10. Urethral mucosal prolapse occurs in postmenopausal women and prepubertal girls.

11. A distinct layer of periurethral fascia should be preserved in managing excision of UDs for reconstruction.
12. When cancer occurs in a bladder diverticulum, the lack of a defined muscle wall makes biopsy and staging difficult because a deep biopsy may cause perforation; without a deep biopsy, proper staging may not be possible.
13. A UD may extend partially around the urethra, anterior to the urethra, or circumferentially around the urethra.
14. Stress incontinence occurs in approximately 10% of women after a repair of a UD.
15. Congenital bladder diverticula are usually solitary, often large, and not associated with bladder outlet obstruction.
16. The ostium (opening) of a UD is most commonly found in the middle or distal thirds of the urethra, at the 4 and 8 o'clock positions.

131

# Surgical Procedures for Sphincteric Incontinence in the Male

*Hunter Wessells, and Alex J. Vanni*

## Keywords

artificial sphincter; male sling; male stress incontinence; post prostatectomy; surgery urethra

## Questions

1. Male stress urinary incontinence (SUI) may result from all of the following except:
    a. radical prostatectomy
    b. pelvic fracture urethral injury
    c. spinal cord injury
    d. bladder extrophy
    e. multiple system atrophy
2. The Advance male sling is hypothesized to provide continence by:
    a. mucosal coaptation.
    b. elevation of the bulbar urethra.
    c. compression of the membranous urethra.
    d. circumferential occlusion of the bulbar urethra.
    e. repositioning and lengthening of the membranous urethra.
3. Nerve-sparing radical prostatectomy has been associated with which of the following:

a. Earlier return to urinary continence and improved final urinary continence
b. Delayed return to urinary continence
c. Earlier return of urinary continence
d. Improved final continence
e. No difference in time to return of continence

4. The initial evaluation of a man with SUI should include all of the following except:
   a. Detailed history and physical exam
   b. Voiding diary and pad test
   c. Post void residual (PVR)
   d. Urinalysis
   e. Pressure flow urodynamics
5. A 65-year-old man status post robotic assisted radical prostatectomy had a direct vision internal urethrotomy for a vesicourethral anastomotic stenosis 6 months ago. He has SUI, wears 6 pads per day, and does not leak at night. Office examination reveals a normal urinalysis, PVR of 175 mL, and positive leak test on Valsalva. What is the next best step?
   a. Cystoscopy
   b. Retrograde urethrogram
   c. Pressure flow urodynamics
   d. Advance sling
   e. Artificial urinary sphincter (AUS)
6. Recurrent moderate SUI 4 years following AUS placement is most likely due to:
   a. detrusor overactivity.
   b. bladder neck stenosis.
   c. subcuff atrophy.
   d. pressure regulating balloon (PRB) herniation.
   e. cuff erosion.
7. A 71-year-old male underwent successful implantation of an Advance sling for stress incontinence. Preoperatively he wore two pads per day, and 1 week postoperatively he is perfectly continent. Two weeks postoperatively he went on a hike with his family and immediately noticed that he was requiring two pads per day again.

He is now 3 weeks postop. What is the best option to treat his recurrent incontinence?

a. Immediate re-exploration, removal of the sling, and replacement with another Advance sling
b. Placement of a second Advance sling in 3 months
c. AUS in 3 months
d. Urethral bulking agent
e. Removal of the sling in 3 months and placement of a Virtue sling

8. A 57-year-old man presents with SUI 12 months after robot-assisted radical prostatectomy and adjuvant external beam radiation therapy. His PSA is <0.01. His 24 hour pad weight is 600 g, and he has incontinence while sleeping. What is his best option for maximal urinary control?

a. Advance male sling
b. Virtue male sling
c. Urethral bulking agent
d. AUS
e. Sacral neuromodulation

9. A 74-year-old man has SUI after radical prostatectomy with a 24-hour pad weight of 600 g. He would like treatment of his SUI and is considered an acceptable surgical candidate. He has poor manual dexterity. Which treatment is his best option for urinary control?

a. Advance sling
b. Virtue sling
c. Urethral bulking agent
d. AUS
e. External collecting device

10. A 72-year-old man with incontinence after radical prostatectomy and adjuvant radiation therapy had an AUS implanted 5 years ago and has recurrent incontinence due to subcuff atrophy. He is impotent and not interested in sexual activity. Intraoperatively, the urethra measures <3.5 cm under the previous cuff and the bulbar urethra distal to this appears atrophic as well and likely measures <4.0 cm. What is the next best step?

a. Place a second cuff in tandem distal to the existing cuff.

b. Place a transcorporal cuff distal to the existing cuff site.
c. Place a transcorporal cuff at the existing cuff site.
d. Place a 3.5-cm cuff distal to the existing cuff site.
e. Place a 3.5-cm cuff at the existing cuff site.

11. A 63-year-old man underwent uncomplicated 4.0-cm cuff bulbar AUS placement and has failed a series of voiding trials over the first week postoperatively. What is the best option to manage his bladder?
    a. Suprapubic cystostomy drainage
    b. 12 French catheter for an additional week followed by another voiding trial
    c. Clean intermittent catheterization (CIC) until able to spontaneously void
    d. Re-exploration and upsize of AUS cuff
    e. Re-exploration and removal of AUS
12. Artificial urinary sphincter infections are most commonly due to which organisms?
    a. *Escherichia coli*
    b. Proteus
    c. *Streptococcus*
    d. *Staphylococcus*
    e. Yeast
13. All of the following are established risk factor for AUS erosion except:
    a. Prior radiation
    b. Urethral catheterization
    c. Prior AUS revision
    d. Prior urethroplasty
    e. Prior Advance sling
14. Management of AUS cuff atrophy can be performed in which way?
    a. Cuff downsizing at existing cuff location
    b. Repositioning a cuff at a different location
    c. Tandem cuff placement
    d. Transcorporal cuff placement
    e. All of the above

15. A 67-year-old man status who underwent implantation of a 3.5-cm cuff bulbar AUS for post–radical prostatectomy SUI develops a symptomatic vesicourethral anastomotic stenosis. How is this best managed?
    a. Urethral dilation with sounds
    b. Antegrade incision through a SP cystostomy tract
    c. Transurethral incision through a 21 Fr rigid cystoscope
    d. Surgically uncouple the cuff followed by endoscopic treatment
    e. Robotic or transperineal bladder neck reconstruction

## Answers

1. **c. Spinal cord injury.** SUI develops only in men with concomitant internal and external sphincter impairment. Internal sphincter incompetence results from pelvic surgery, such as radical prostatectomy, bladder neck injury, specific sympathetic neuropathic dysfunction, or embryological disruption. Incompetence of the external sphincter occurs most frequently after radical prostatectomy, but also can result from pelvic fracture urethral injuries, traumatic and acquired myelopathy, and congenital disorders such as spinal dysraphism, sacral agenesis, and the exstrophy/epispadias complex. Spinal cord injury leads to external or internal sphincter dyssynergia, which cause failure to empty, not failure to store.
2. **e. Repositioning and lengthening of the membranous urethra. Transobturator slings are hypothesized to enhance external sphincter function by repositioning and lengthening the membranous urethra without significant compression.** Urodynamic studies show an increase in abdominal leak point pressure without other urodynamic evidence of obstruction. Evidence for this comes from fluoroscopic studies and magnetic resonance imaging. The latter studies demonstrate an increase in membranous urethral length and elevation of the bladder neck, posterior bladder wall, and rhabdosphincter.

3. **c. Earlier return of urinary continence.** In systematic review and meta-analysis, nerve-sparing techniques have been associated with **more rapid return of continence, although not final continence status, after RP**.
4. **e. Pressure flow urodynamics.** The initial evaluation of a man with UI requires a detailed history, physical examination, postvoid residual measurement, and urinalysis. **Pressure flow urodynamics** can be used to evaluate potential bladder neck stenosis, residual sphincteric function, and bladder storage function as indicated, but **is not part of the initial evaluation of stress incontinence.**
5. **a. Cystoscopy.** Because unrecognized urethral or bladder neck pathology can significantly complicate all surgical approaches, endoscopic evaluation should be considered prior to surgical correction of UI post RP or TURP if there is any suspicion of bladder neck stenosis. **Cystoscopy is the best way to evaluate his elevated PVR and determine if he has a recurrent vesicourethral anastomotic stenosis.** Urethrography is less useful in delineating posterior urethral strictures. Urodynamics are not indicated unless poor bladder emptying cannot be otherwise explained.
6. **c. Subcuff atrophy.** While detrusor overactivity, PRB herniation and cuff erosion may all result in recurrent UI, **subcuff atrophy is the most common cause of recurrent SUI 4 years after AUS placement.** Mechanical failure, another cause of recurrent incontinence, is not listed as an option but over time becomes more common.
7. **a. Immediate re-exploration, removal of the sling, and replacement with another Advance sling.**
   **Evaluation of the patient who had prior urethral sling is similar to the evaluation after AUS placement** with several caveats. Identifying the type of sling placed (Advance vs. Virtue vs. bone anchored) and the timing of the incontinence are important. In the case of the Advance sling, it is important to determine whether the patient may have loosened the sling in the first 6 weeks postoperatively, whether the incontinence is persistent (ongoing SUI) or the result of detrusor overactivity. If patients give a history

of improved UI and a loosening of the sling in the first 6 weeks postoperatively, re-exploration is appropriate. The perineum is opened, the arms of the sling lateral to the bulbar urethra are identified and each arm is cut. The remaining central portion of the sling should be removed from the surface of the corpus spongiosum of the bulbar urethra. Another sling can then be placed in the same manner, as described later.

8. **d. AUS.** Because of limited efficacy, submucosal bulking agents are not part of the treatment algorithm for post-RP UI. Thus the **AUS and slings should be considered as first-line surgical therapy for sphincteric incontinence in most men, although a trial of a bulking agent may be appropriate in cases of neurogenic male stress UI. The AUS remains the established device for treatment of moderate to severe UI, supported by numerous publications documenting its benefits.** Thus, the transobturator slings should be primarily used in cases with mild to moderate incontinence, which can be defined as a 24-hour pad weight of less than 150 grams for mild UI and less than 400 grams for moderate UI. A sling procedure should be offered with caution to those with prior radiation therapy or urethral erosion, because the degree of urine loss that exists in this group usually exceeds the limits of the procedure. One reason for the lower efficacy in these patients may be the lack of appropriate tissue elasticity for deformation/elongation/compression of the external urinary sphincter apparatus.
9. **a. Advance sling.** Because of limited efficacy, submucosal bulking agents are not part of the treatment algorithm for post-RP UI. Although the patient has severe UI, an **AUS is not advised due to his poor manual dexterity, and a perineal sling is the preferred treatment option.** The authors recommend the Advance sling based on broader experience with this device in the urological community, more substantial corroborating literature, and personal experience with the device compared to the Virtue sling.
10. **b. Place a transcorporal cuff distal to the existing cuff site.** Indiscriminate use of the tandem cuff approach as first-line surgical treatment should be tempered by higher rates of erosion associated

with the distal cuff in several series. **Use of the 3.5-cm size has resulted in an increased risk of erosion in patients with a prior history of radiation therapy.** Patients with radiation therapy or other adverse risk factors and significant urethral atrophy should be considered for transcorporal placement rather than a 3.5-cm cuff. Placing a transcorporal cuff should be performed at a different location along the bulbar urethra.

11. **a. Suprapubic cystostomy drainage. If the patient fails a voiding trial at 48 hours, suprapubic cystostomy drainage is recommended to reduce the risk of urethral erosion. Ultrasound or fluoroscopic guidance is recommended to prevent puncture or potential contamination of the PRB.** Retention persisting beyond several weeks implies undersizing of the cuff; in such cases reoperation and cuff replacement may be required. **Prolonged urethral catheterization or CIC after AUS will result in an increased risk of cuff urethral erosion, even if the device is deactivated.**
12. d. ***Staphylococcus.*** **Skin pathogens are the most commonly cultured organism, usually *Staphylococcus epidermidis*** and *Staphylococcus aureus*.
13. **e. Prior Advance sling.** Prior radiation, urethral catheterization, prior AUS revision, and urethroplasty are all risk factors for AUS urethral erosion. **Prior Advance sling has not resulted in an increased risk of AUS erosion.**
14. **e. All of the above. Managing AUS cuff atrophy can be performed in a number of ways and may require any one of the following maneuvers—cuff downsizing at existing cuff location, repositioning a cuff at a different location, tandem cuff placement, or transcorporal cuff placement—depending** on intraoperative findings.
15. d. **Surgically uncouple the cuff followed by endoscopic treatment.** First-line management of vesicourethral anastomotic strictures usually involves an endoscopic approach with DVIU or dilation. **In the event significant endoscopic manipulation is required proximal to the cuff site, our practice is to surgically uncouple the cuff** for the duration of the endoscopic procedure to

avoid the risk of urethral erosion or damaging the AUS cuff, which may result from either a balloon dilation or transurethral incision through a 21-Fr rigid cystoscope.

## Chapter Review

1. A careful history and voiding diary characterize the type and severity of urinary incontinence in most men.
2. Uroflometry and post void residual urine volume measurement should be assessed at minimum.
3. Suspected outlet obstruction, significant bladder overactivity, or impaired bladder contractility should prompt urodynamic studies.
4. Patients with suspected urethral, vesicourethral, or bladder neck pathology should undergo cystoscopy.
5. Urinalysis (and culture, if indicated) is required prior to surgical treatment.
6. Surgical intervention is indicated for SUI due to ISD that fails to improve with conservative management.
7. Implant selection depends on the severity of UI, prior surgical procedures, bladder function, cystoscopic findings, manual dexterity and cognitive function, long-term risk of complications and reoperation, and patient preference.
8. Slings should only be used very selectively in cases of radiation and severe gravitational urinary incontinence.
9. AUS should not be used in cases of poor manual dexterity, cognitive disability, poor urethral tissue integrity, or severely reduced bladder compliance.
10. Standard implantation of the bulbar AUS places the cuff around the corpus spongiosum proximal to the convergence of the corporal bodies.
11. Pressure regulating balloon of 61 to 70 cm $H_2O$ is advised for bulbar urethral cuffs; placement may be via abdominal, scrotal, or perineal approach.
12. Postoperative deactivation of the cuff for 4–6 weeks is essential for proper healing without erosion.

13. Prolonged urinary retention requires suprapubic cystostomy drainage.
14. Revision surgery for nonmechanical causes may require transcorporal cuff placement or other modifications such as using 3.5-cm or tandem cuff.
15. Male bulbar urethral slings are indicated in men with mild to moderate sphincteric UI, intact bladder contractility, and well-vascularized nonradiated urethral anatomy.
16. Transobturator slings reposition and elongate the membranous urethra and depend on the intact external sphincter.
17. Quadratic fixation provides an alternative to the two arm devices with limited data on efficacy and durability.
18. Device infection typically requires total explantation, but salvage procedures in select patients with device removal, antibiotic regimen washout, and reimplantation are feasible.
19. Device erosion requires total explantation and catheter drainage.
20. Troubleshooting of recurrent UI post AUS requires a systematic and exhaustive evaluation.
21. Cuff atrophy requires downsizing, repositioning, tandem cuff, or transcorporal surgery.
22. Artificial urinary sphincter continues to provide a reliable solution to male UI of all degrees of severity across the range of first-time implantation to complex reoperative and high-risk patients.
23. Transobturator slings such as Advance and newer devices provide long-term durable outcomes for mild to moderate male UI after radical prostatectomy.

# PART XIV

# Benign and Malignant Bladder Disorders

## OUTLINE


132. Bladder Surgery for Benign Disease

133. Genital and Lower Urinary Tract Trauma

134. Special Urologic Considerations in Transgender Individuals

135. Tumors of the Bladder

136. Management Strategies for Non–Muscle-Invasive Bladder Cancer (Ta, T1, and CIS)

137. Management of Muscle-Invasive and Metastatic Bladder Cancer

138. Surgical Management of Bladder Cancer: Transurethral, Open, and Robotic

139. Use of Intestinal Segments in Urinary Diversion

140. Cutaneous Continent Urinary Diversion

141. Orthotopic Urinary Diversion

142. Minimally Invasive Urinary Diversion

132

# Bladder Surgery for Benign Disease

*Paras H. Shah, and Lee Richstone*

## Keywords

benign disease; bladder surgery; distal ureter; urinary bladder

## Questions

1. All of the following are critical to consider in the surgical management of a bladder diverticulum EXCEPT one. Which?
    a. If outlet obstruction is present, it should be managed prior to or at the time of bladder diverticulectomy.
    b. The need for ureteral reimplantation is a contraindication to surgical resection of a bladder diverticulum.
    c. Cystoscopic examination of the diverticulum should be attempted in order to characterize the anatomy of the diverticulum as well as to evaluate for the presence for a bladder tumor within the diverticulum.
    d. A preoperative stent may be placed if there is concern for ureteral injury during bladder diverticulectomy.
2. In patients with ureteral stricture disease, which of the following should ideally be a part of the preoperative evaluation?
    a. Retrograde pyelography
    b. Both retrograde and antegrade evaluation
    c. Computed tomography (CT) urography

d. None of the above; direct intraoperative visualization of stricture length will suffice.

3. Which of the following maneuvers can be used to reduce tension on the ureterovesical anastomosis?
    a. Ligation of the contralateral bladder pedicles to facilitate cephalad displacement of the bladder
    b. Disengagement of the peritoneal reflection off the bladder dome to enhance bladder mobility
    c. A psoas hitch procedure
    d. Renal mobilization and downward nephropexy
    e. All of the above
4. All of the following are essential surgical aspects of the Boari flap or bladder advancement flap EXCEPT one. Which?
    a. An adequate-sized bladder must be present (200- to 300-mL capacity).
    b. The contralateral vesical pedicle may be transected if necessary.
    c. The bladder flap should be slightly shorter than anticipated because bladder tissue can easily be stretched.
    d. A tension-free anastomosis is important.
    e. Typically, a refluxing ureteral anastomosis is created.
5. What nerves may be injured during a psoas hitch procedure?
    a. Obturator nerve
    b. Genitofemoral nerve
    c. Ilioinguinal nerve
    d. Sciatic nerve
6. Which of the following statements is NOT correct regarding enterocystoplasty?
    a. Subtotal cystectomy is not always mandatory.
    b. The mesenteric pedicle of the selected bowel segment is wide and broad-based.
    c. The mesenteric window is closed.
    d. Reestablishment of bowel continuity is a critical step of the operation and may be performed extracorporeally, if necessary, for added security.

e. Bowel-to-bladder anastomosis is optimally performed with interrupted serosa-to-serosa sutures.

7. Contraindications for enterocystoplasty include all of the following EXCEPT:
   a. diverticulosis.
   b. inflammatory bowel disease.
   c. renal failure.
   d. noncompliance.
8. Which is the following is a contraindication to partial cystectomy for a bladder leiomyoma?
   a. Need for ureteral reimplantation
   b. Transmural involvement by the neoplastic process
   c. Significant irritative voiding symptoms at baseline
   d. Reduced preoperative bladder capacity
   e. c and d
9. Which of the following with regard to ureteral reimplantation is true?
   a. A nonrefluxing anastomosis is superior to a refluxing anastomosis due to a reduced risk for infection.
   b. The ureter should be skeletonized as much as possible to facilitate maximal ureteral mobility and increase the chance for a tension-free anastomosis.
   c. For a nonrefluxing anastomosis, the length of the submucosal tunnel should be approximately three to five times the diameter of the ureter to be reimplanted.
   d. For ureteral defects greater than 4 cm, a primary ureteroneocystotomy is generally feasible without the need for a psoas hitch or Boari flap.
10. When urachal surgery is performed for a benign indication, which of the following is true?
   a. Wide margins of resection along the peritoneal wing are necessary.
   b. A wide-margin bladder cuff is preferred.
   c. Excision of the umbilicus should be performed.
   d. Narrow peritoneal wings around the urachus and a narrow-margin bladder cuff will suffice.

# Answers

1. **b. The need for ureteral reimplantation is a contraindication to surgical resection of a bladder diverticulum.** The need for ureteral reimplantation is not a contraindication to performing bladder diverticulectomy. Rather, careful preoperative planning is necessary to discern the proximity of the ureter to the diverticulum and to gauge whether surgical maneuvers, such as retrograde ureteral stent placement, are necessary to prevent inadvertent ureteral injury. Alternatively, planned ureteral reimplantation may be performed at the time of bladder diverticulectomy. The surgeon should be cognizant of the ureteral anatomy relative to the diverticulum to avoid intraoperative injury. At the conclusion of the case, the ureter should be inspected to ensure its integrity; reimplantation should be performed if there is any concern for ureteral compromise.
2. **b. Both retrograde and antegrade evaluation.** Both antegrade and retrograde evaluation of stricture disease is necessary to accurately assess the length and location of disease as well as whether multiple strictures are present, as these factors heavily influence operative planning. In cases of distal ureteral strictures, retrograde contrast administration alone may fail to opacify more proximal aspects of the ureter, precluding an understanding of the length of stricture disease. Similarly, high-grade proximal strictures may prevent adequate ureteral opacification past the stricture segment with antegrade studies (e.g., contrast administration through a nephrostomy tube or delayed-phase imaging on a CT urogram), hindering adequate characterization of the extent of stricture disease. Obstructive uropathy in the setting of a ureteral stricture may also impair the uptake of intravenous contrast by the kidney during a CT urogram study, rendering this study futile for the anatomic characterization of stricture disease when it is performed in isolation.
3. **e. All of the above.** It is critical to avoid tension at the ureterovesical junction during reimplantation as ischemic injury

related to undue traction may result in the formation of scar and subsequent stricture. All the aforementioned techniques can be employed to reduce tension, although they vary in their degree of surgical complexity. In general, a psoas hitch procedure represents the least complex of the options mentioned and is generally all that is needed for ureteral strictures involving the distal third of the ureter. If further length is needed, a Boari flap may be performed prior to more aggressive surgical maneuvers such as ligation of bladder pedicles, disengagement of the peritoneal reflection, and renal mobilization.

4. **c. The bladder flap should be slightly shorter than anticipated because bladder tissue can easily be stretched.** The bladder flap should be slightly shorter than anticipated because bladder tissue can easily be stretched. A tension-free anastomosis of the anterolateral bladder flap based on the ipsilateral vesical pedicle is critical. The bladder flap should be somewhat longer and wider than anticipated because the nondistended bladder shrinks in size, thus placing tension on the anastomosis. Ideally, a flap length-to-breadth ratio of 3:1 ensures good vascularity of its apex.
5. **b. Genitofemoral nerve.** During a psoas hitch procedure, sutures should be placed superficially through the tendon of the psoas minor parallel to the direction of the muscle fibers. This helps to minimize the risk for disrupting the genitofemoral nerve, situated superficially on the psoas muscle, or the femoral nerve, located deeper in the belly of the muscle.
6. **e. Bowel-to-bladder anastomosis is optimally performed with interrupted serosa-to-serosa sutures.** The technical principles of enterocystoplasty are identical between open surgical and laparoscopic techniques. Generous mobilization of the bladder allows creation of an adequate anteroposterior cystotomy. Subtotal cystectomy is necessary only in patients with severely symptomatic interstitial cystitis. An optimal segment of bowel based on a broad, well-vascularized mesenteric pedicle is selected that will reach the pelvis without tension. The bowel segment is isolated and bowel continuity reestablished, either by intracorporeal or extracorporeal techniques, and the mesenteric window is closed. The isolated

bowel segment is detubularized and a bowel plate is created appropriately. A tension-free, watertight, full-thickness circumferential running anastomosis of the bowel segment to the bladder is created. Adequate urinary drainage is established.

7. **a. Diverticulosis.** The presence of bowel pathology such as diverticulitis or ulcerative colitis requires the use of alternative, nondiseased bowel segments. Enterocystoplasty should not be performed in the presence of advanced renal or liver failure, inflammatory bowel disease, or short gut syndrome or in a patient who is unable to perform or noncompliant in performing intermittent catheterization reliably. Diverticulosis is not a contraindication for performing enterocystoplasty.
8. **e. c and d.** Bladder leiomyomas are benign tumors of the bladder that most commonly manifest with an endovesical, intramural, or extravesical pattern of growth; therefore they may not be discerned on cystoscopic exam or may appear only as a submucosal growth. Owing to the benign nature of bladder leiomyomas, partial cystectomy represents a reasonable treatment approach, particularly for lesions that are large in size and require wide local excision. Patient selection is critical when partial cystectomy is being considered irrespective of the pathology for which it is being performed. This includes performing the procedure only in patients with adequate preoperative bladder capacity, as further postoperative reduction in capacity can result in significant voiding complaints. Additionally, the presence of baseline irritative voiding complaints is a relative contraindication, as the process of bladder tissue resection and postoperative reduction in bladder capacity may exacerbate these symptoms. Ureteral reimplantation is not an absolute contraindication for partial cystectomy, particularly for benign disease, and can be performed concomitantly if necessary.
9. **c. For a nonrefluxing anastomosis, the length of the submucosal tunnel should be approximately three to five times the diameter of the ureter to be reimplanted.** When a nonrefluxing anastomosis is being performed, a submucosal tunnel is created to support compression of the reimplanted ureter during bladder contraction. The length of this tunnel should be roughly three to five times the

diameter of the ureter to be reimplanted. Nonrefluxing anastomoses can be created via an extravesical or transvesical approach. Based on the currently available literature, the use of a refluxing anastomosis does not appear to place individuals at increased risk for accelerated renal functional decline or for stenosis as compared with nonrefluxing anastomoses. Similarly, long-term studies in adults undergoing a refluxing anastomosis do not seem to indicate increased risk for infection. For ureteral defects greater than 4 cm, primary ureteroneocystotomy would likely result in undue traction at the ureterovesical anastomosis and cause stricture secondary to ischemic injury. Therefore additional measures such as psoas hitch and/or Boari flap are often implemented. In addition to creating a tension-free anastomosis during ureteral reimplantation, the ureter should be manipulated delicately so as to minimize devascularization injury. Furthermore, a conscious effort should be made during extravesical mobilization of the ureter to preserve the periureteral fibroadipose tissue and avoid excessive skeletonization, which may also compromise the blood supply.

10. **d. Narrow peritoneal wings around the urachus and a narrow-margin bladder cuff will suffice.** Surgery is indicated for symptomatic or infected urachal anomalies that do not respond to conservative approaches to treatment, such as incision and drainage and/or antibiotic therapy. When surgery is performed, the entire urachus should be excised, including a small area on the dome of the bladder. The practice of incorporating a bladder cuff into the surgical specimen is thought to ensure that no residual tissue will be left in the vicinity of the bladder apex, as this may risk recurrence of symptoms or malignant transformation. In general, a narrow peritoneal wing around the urachus will suffice, as will a narrow bladder cuff because of the noninfiltrative nature of benign urachal lesions. Excision of the umbilicus is not necessary for a benign indication, especially when the lesion is not in the vicinity of the umbilicus.

## Chapter Review

1. For bladder diverticulectomy, the success of surgery is contingent on a meticulous preoperative workup that characterizes the number, size, and location of diverticula. Specifically, the anatomy relative to the ureter should be defined to ensure preservation of the ureter's integrity during repair.
2. The disease process underlying formation of the bladder diverticulum should be managed prior to or at the time of diverticular repair.
3. The fibrous pseudocapsule surrounding the diverticulum serves as a cleavage plane to free the diverticulum from surrounding extravesical tissue.
4. During minimally invasive bladder diverticulectomy, selective catheterization of the diverticulum with a Council-tip catheter placed over a guidewire allows for selective filling of the diverticulum and aids in intraoperative identification.
5. In ureteral reimplantation, aggressive skeletonizing of the ureter during mobilization should be avoided, as this may lead to ischemia and ureteral stricture formation.
6. A tension-free anastomosis between the bladder and ureter is critical, as the presence of tension risks anastomotic stricture formation.
7. The ureter should be adequately spatulated prior to creation of the anastomosis.
8. When a nonrefluxing ureteral reimplant is being performed, the length of the submucosal tunnel should be three to five times the caliber of the ureteral lumen.
9. Longer ureteral defects may be bridged through maneuvers such as ligation of the contralateral medial umbilical ligament, dissection of the peritoneal reflection off the bladder dome, or ligation of the contralateral bladder pedicle, all of which are conducive to cephalad displacement of the bladder.
10. During creation of a psoas hitch, sutures that fix the extravesical surface of the bladder to the psoas minor tendon should be placed superficially and parallel to the direction of the muscle fibers to avoid entrapment of the femoral and genitofemoral nerves, respectively.
11. Reimplantation of the ureter should be performed prior to tying the anchoring sutures to the psoas muscle tendon during psoas hitch

creation.

12. The base of a Boari flap should measure at least 4 cm in width, with longer flaps requiring wider base measurement to maintain adequate blood supply; in this regard, the length-to-base ratio should not exceed 3:1.
13. If a non-refluxing anastomosis is planned into a Boari flap, there should be at least 3 cm of overlap between the flap and the ureteral segment.
14. Due to its relative simplicity and versality, advancement flaps may be used in lieu of Boari flaps to bridge longer ureteral defects.
15. In enterocystoplasty, the ileum is most frequently utilized for bladder augmentation given its abundance, ease of mobilization, and robust blood supply.
16. The ileum should be isolated approximately 20 cm proximal to the ileocecal valve to spare the terminal ileum and avoid risk for bile salt and vitamin B12 malabsorption.
17. Detubularization of the isolated intestinal segment is necessary to reduce coordinated peristaltic contractions, which may otherwise contribute to high intravesical pressures
18. A broad mesenteric base to the isolated intestinal segment should be maintained to avoid torsion of the pedicle as well as preserve a rich blood supply.
19. The bladder should be generously bivalved along its dome so as to facilitate a wide-mouth anastomosis with the intestinal patch; a narrow-mouth anastomosis would result in the creation of a pseudo-diverticulum that may poorly empty and cause complications.
20. Patients should be counseled regarding the importance of continuous intermittent catheterization as the failure to do so risks perforation.
21. Preoperative evaluation should confirm the presence of adequate bladder capacity and the absence of significant voiding symptoms prior to proceeding with partial cystectomy.
22. Partial cystectomy for benign disease processes still require adequate margins of resection given the infiltrative nature of many conditions and the propensity for recurrence in cases of inadequate excision.
23. During minimally invasive partial cystectomy, intraoperative flexible cystoscopy by the bedside assistant can aid the surgeon in

identification of the site of bladder pathology.

24. Urachal surgery, when being performed for benign indication, does not require a wide margin of resection of the bladder dome around the site of urachal insertion.
25. Approach to bladder stone surgery depends on the size and number of calculi as well as the presence of any associated processes contributing to stone formation that may need to be simultaneously addressed.
26. Percutaneous cystolithotripsy is a minimally invasive alterative to management of large bladder calculi; percutaneous access, however, may be prohibited in patients with prior pelvic or abdominal surgery given concern over intestinal injury.